Panel IV Results − Specification 1: Wage only

Treatment: real AEWR (2012 dollars). County and year fixed effects; standard errors clustered by AEWR−region x target subregion.

Donor wage level: county−mapped OEWS−area Big−Six hourly−wage proxy; QCEW supplies county employment and calibration features; source year t−1.

	H−2A outcomes						Farm outcomes					
Statistic	**Certified workers / 2011 farm emp.**	**Certified hours / 2011 farm emp.**	**Applications / 2011 farm employment**	**Linked worksite businesses / 2011 farm emp.**	**Certified positions / application**	**Certified hours / position**	**Real crop price index**	**Farm employment**	**Production expense share**	**Real farm income / worker**	**Hired−labor expense share**	**Crop output quantity index**
AEWR coefficient	**−0.0966***	**−57.80***	**−0.0014**	**−0.0022**	**0.7453**	**−117.0****	**−4.206**	**−8.666**	**−0.0272**	**36,067****	**0.0064****	**−7.442**
Clustered standard error	(0.0538)	(34.37)	(9.17e−04)	(0.0034)	(3.374)	(53.02)	(6.584)	(10.54)	(0.0302)	(14,095)	(0.0031)	(136.5)
p−value	0.076	0.097	0.124	0.510	0.826	0.030	0.525	0.414	0.370	0.012	0.040	0.957
First−stage excluded−instrument F	22.90	22.90	22.90	22.90	19.35	19.24	22.85	22.90	22.90	22.90	22.90	22.85
Observations	33,497	33,497	33,497	33,497	21,118	20,825	33,441	33,497	33,497	33,497	33,497	33,441
Counties	2,848	2,848	2,848	2,848	2,553	2,538	2,847	2,848	2,848	2,848	2,848	2,847
Inference clusters	80	80	80	80	80	80	80	80	80	80	80	80

Notes: Coefficients report the effect of a one−dollar increase in real AEWR. Clustered standard errors are in parentheses; * p<0.10, ** p<0.05, *** p<0.01. The first−stage F is the squared clustered t statistic for the excluded instrument.

Panel IV Results − Specification 2: Seasonal/composition

Treatment: real AEWR (2012 dollars). County and year fixed effects; standard errors clustered by AEWR−region x target subregion.

Donor wage level: county−mapped OEWS−area Big−Six hourly−wage proxy; QCEW supplies county employment and calibration features; source year t−1.

	H−2A outcomes						Farm outcomes					
Statistic	**Certified workers / 2011 farm emp.**	**Certified hours / 2011 farm emp.**	**Applications / 2011 farm employment**	**Linked worksite businesses / 2011 farm emp.**	**Certified positions / application**	**Certified hours / position**	**Real crop price index**	**Farm employment**	**Production expense share**	**Real farm income / worker**	**Hired−labor expense share**	**Crop output quantity index**
AEWR coefficient	**−0.0849***	**−54.32***	**−0.0022**	**−0.0037**	**0.7336**	**−87.73***	**−2.642**	**−13.15**	**−7.75e−04**	**36,079*** **	**0.0056** **	**−3.465**
Clustered standard error	(0.0477)	(29.84)	(0.0013)	(0.0037)	(2.881)	(47.02)	(5.923)	(10.31)	(0.0297)	(12,840)	(0.0028)	(118.1)
p−value	0.079	0.073	0.109	0.313	0.800	0.066	0.657	0.206	0.979	0.006	0.050	0.977
First−stage excluded−instrument F	39.45	39.45	39.45	39.45	30.92	30.85	39.33	39.45	39.45	39.45	39.45	39.33
Observations	33,497	33,497	33,497	33,497	21,118	20,825	33,441	33,497	33,497	33,497	33,497	33,441
Counties	2,848	2,848	2,848	2,848	2,553	2,538	2,847	2,848	2,848	2,848	2,848	2,847
Inference clusters	80	80	80	80	80	80	80	80	80	80	80	80

Notes: Coefficients report the effect of a one−dollar increase in real AEWR. Clustered standard errors are in parentheses; * p<0.10, ** p<0.05, *** p<0.01. The first−stage F is the squared clustered t statistic for the excluded instrument.

Panel IV Results − Specification 3: Wage only + controls

Treatment: real AEWR (2012 dollars). County and year fixed effects; standard errors clustered by AEWR−region x target subregion.

Donor wage level: county−mapped OEWS−area Big−Six hourly−wage proxy; QCEW supplies county employment and calibration features; source year t−1.

	H−2A outcomes						Farm outcomes					
Statistic	**Certified workers / 2011 farm emp.**	**Certified hours / 2011 farm emp.**	**Applications / 2011 farm employment**	**Linked worksite businesses / 2011 farm emp.**	**Certified positions / application**	**Certified hours / position**	**Real crop price index**	**Farm employment**	**Production expense share**	**Real farm income / worker**	**Hired−labor expense share**	**Crop output quantity index**
AEWR coefficient	**−0.1102***	**−70.16***	**−0.0011**	**−8.58e−04**	**−1.907**	**−115.0****	**−4.379**	**−4.828**	**−0.0249**	**37,550****	**0.0065***	**13.79**
Clustered standard error	(0.0564)	(36.82)	(8.98e−04)	(0.0031)	(3.427)	(57.12)	(6.722)	(9.794)	(0.0336)	(14,382)	(0.0033)	(177.6)
p−value	0.054	0.060	0.225	0.784	0.580	0.047	0.517	0.623	0.460	0.011	0.050	0.938
First−stage excluded−instrument F	24.79	24.79	24.79	24.79	20.26	19.98	24.84	24.79	24.79	24.79	24.79	24.84
Observations	33,497	33,497	33,497	33,497	21,118	20,825	33,441	33,497	33,497	33,497	33,497	33,441
Counties	2,848	2,848	2,848	2,848	2,553	2,538	2,847	2,848	2,848	2,848	2,848	2,847
Inference clusters	80	80	80	80	80	80	80	80	80	80	80	80

Notes: Coefficients report the effect of a one−dollar increase in real AEWR. Clustered standard errors are in parentheses; * p<0.10, ** p<0.05, *** p<0.01. The first−stage F is the squared clustered t statistic for the excluded instrument.

Panel IV Results − Specification 4: Seasonal/composition + controls (preferred)

Treatment: real AEWR (2012 dollars). County and year fixed effects; standard errors clustered by AEWR−region x target subregion.

Donor wage level: county−mapped OEWS−area Big−Six hourly−wage proxy; QCEW supplies county employment and calibration features; source year t−1.

	H−2A outcomes						Farm outcomes					
Statistic	**Certified workers / 2011 farm emp.**	**Certified hours / 2011 farm emp.**	**Applications / 2011 farm employment**	**Linked worksite businesses / 2011 farm emp.**	**Certified positions / application**	**Certified hours / position**	**Real crop price index**	**Farm employment**	**Production expense share**	**Real farm income / worker**	**Hired−labor expense share**	**Crop output quantity index**
AEWR coefficient	**−0.0986****	**−66.95****	**−0.0019**	**−0.0022**	**−2.002**	**−80.94**	**−3.092**	**−9.993**	**0.0018**	**38,509*****	**0.0055***	**16.45**
Clustered standard error	(0.0489)	(32.46)	(0.0013)	(0.0035)	(2.881)	(51.77)	(5.875)	(9.325)	(0.0316)	(12,978)	(0.0029)	(156.1)
p−value	0.047	0.042	0.162	0.532	0.489	0.122	0.600	0.287	0.954	0.004	0.062	0.916
First−stage excluded−instrument F	46.06	46.06	46.06	46.06	34.18	33.86	46.17	46.06	46.06	46.06	46.06	46.17
Observations	33,497	33,497	33,497	33,497	21,118	20,825	33,441	33,497	33,497	33,497	33,497	33,441
Counties	2,848	2,848	2,848	2,848	2,553	2,538	2,847	2,848	2,848	2,848	2,848	2,847
Inference clusters	80	80	80	80	80	80	80	80	80	80	80	80

Notes: Coefficients report the effect of a one−dollar increase in real AEWR. Clustered standard errors are in parentheses; * p<0.10, ** p<0.05, *** p<0.01. The first−stage F is the squared clustered t statistic for the excluded instrument.